

Stochastic Calculus & Option Pricing — Synthetic Notes

Nathan Sauldubois

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How to use these notes.

- These are *synthetic* lecture notes: definitions, key results, and the computations you actually use.
- Each lecture ends with a short checklist: “what you must be able to do”.
- Notation: all processes are defined on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ satisfying the usual conditions.

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1 Lecture 1 — Probability refresher and core mathematical tools

Summary. Random variables and their laws; CDFs and densities; moments, variance, and covariance; dominated convergence; LLN and CLT; characteristic functions; information and conditional expectation; Gaussian conditioning; filtrations and martingales; and the differential-calculus formulas used later.

Reference. S. E. Shreve, *Stochastic Calculus for Finance II*, Chapters 1–2.

1.1 Random variables, laws, CDFs, and densities

Random variable and law. On a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, a real random variable is a measurable map $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Its law is the probability measure

$$\mathbb{P}_X(A) = \mathbb{P}(X \in A), \quad A \in \mathcal{B}(\mathbb{R}).$$

The random variable maps states to numbers; its law records the probabilities of those numbers.

CDF and density. The cumulative distribution function of X is

$$F_X(x) = \mathbb{P}(X \leq x).$$

Every CDF is nondecreasing and right-continuous, with limits 0 at $-\infty$ and 1 at $+\infty$. For $a < b$,

$$\mathbb{P}(a < X \leq b) = F_X(b) - F_X(a).$$

If X has density f_X , then

$$F_X(x) = \int_{-\infty}^x f_X(u) du, \quad \mathbb{P}(a < X \leq b) = \int_a^b f_X(u) du,$$

and $f_X = F'_X$ wherever the CDF is differentiable. A law and a CDF always exist; a density need not exist.

Joint and marginal laws. The joint law of (X, Y) contains both marginal laws and their dependence. In the discrete case,

$$p_X(x) = \sum_y p_{X,Y}(x, y), \quad p_Y(y) = \sum_x p_{X,Y}(x, y),$$

whereas for a joint density,

$$f_X(x) = \int_{\mathbb{R}} f_{X,Y}(x, y) dy, \quad f_Y(y) = \int_{\mathbb{R}} f_{X,Y}(x, y) dx.$$

Independence is equivalent to factorization of the joint law: $p_{X,Y} = p_X p_Y$ or $f_{X,Y} = f_X f_Y$.

Common one-dimensional laws.

Law	Parameters	$\mathbb{E}[X]$	$\text{Var}(X)$
Bernoulli	p	p	$p(1-p)$
Binomial	n, p	np	$np(1-p)$
Poisson	λ	λ	λ
Uniform	$[a, b]$	$(a+b)/2$	$(b-a)^2/12$
Exponential	rate λ	$1/\lambda$	$1/\lambda^2$
Gaussian	μ, σ^2	μ	σ^2

Mass functions and densities to recognize. For the discrete laws,

$$X \sim \text{Bernoulli}(p) : \mathbb{P}(X = x) = p^x(1-p)^{1-x}, \quad x \in \{0, 1\},$$

$$X \sim \text{Binomial}(n, p) : \mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, \dots, n,$$

$$X \sim \text{Poisson}(\lambda) : \mathbb{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k \in \mathbb{N}.$$

For the continuous laws,

$$X \sim \text{Unif}[a, b] : f_X(x) = \frac{1}{b-a} \mathbf{1}_{[a,b]}(x),$$

$$X \sim \text{Exp}(\lambda) : f_X(x) = \lambda e^{-\lambda x} \mathbf{1}_{[0,\infty)}(x),$$

$$X \sim \mathcal{N}(\mu, \sigma^2) : f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

1.2 Expectation, moments, variance, and covariance

Expectation of a transformation. If the relevant sum or integral is finite, then

$$\mathbb{E}[h(X)] = \sum_x h(x)\mathbb{P}(X = x) \quad \text{or} \quad \mathbb{E}[h(X)] = \int_{\mathbb{R}} h(x)f_X(x) \, dx.$$

The k th moment is $\mathbb{E}[X^k]$. Integrability means $\mathbb{E}[|X|] < \infty$, and $\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A)$.

Essential expectation properties. For integrable X, Y and constants a, b ,

$$\begin{aligned} \mathbb{E}[aX + bY] &= a\mathbb{E}[X] + b\mathbb{E}[Y], \\ X \leq Y &\implies \mathbb{E}[X] \leq \mathbb{E}[Y], \\ |\mathbb{E}[X]| &\leq \mathbb{E}[|X|]. \end{aligned}$$

Linearity does not require independence.

Variance and covariance. For square-integrable variables,

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2, \quad \text{SD}(X) = \sqrt{\text{Var}(X)},$$

and

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

The identities used most often are

$$\begin{aligned} \text{Var}(aX + b) &= a^2 \text{Var}(X), \\ \text{Var}(X + Y) &= \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y), \\ \text{Var}\left(\sum_{k=1}^n X_k\right) &= \sum_{k=1}^n \text{Var}(X_k) + 2 \sum_{i < j} \text{Cov}(X_i, X_j). \end{aligned}$$

Independence implies zero covariance, but the converse is false in general. For independent square-integrable variables, the variance of a sum is the sum of the variances.

Theorem 1.1 (Dominated Convergence Theorem). *Suppose $X_n \rightarrow X$ almost surely and there is an integrable random variable Y such that $|X_n| \leq Y$ almost surely for every n . Then X is integrable and*

$$\mathbb{E}[|X_n - X|] \rightarrow 0, \quad \mathbb{E}[X_n] \rightarrow \mathbb{E}[X].$$

Remark 1.2. Dominated convergence is the standard tool for interchanging a limit and an expectation. Pointwise convergence alone is not sufficient; the integrable dominating variable controls the tails uniformly in n .

1.3 Sample moments, LLN, and CLT

For observations $X_1, \dots, X_n$, define

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i, \quad v_n = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2, \quad s_n^2 = \frac{n}{n-1} v_n.$$

Here v_n is the empirical variance and s_n^2 is its usual unbiased version under i.i.d. sampling with finite variance.

Theorem 1.3 (Law of Large Numbers). *If (X_i) are i.i.d. and $\mathbb{E}[|X_1|] < \infty$, then*

$$\overline{X}_n \xrightarrow[n \rightarrow \infty]{\mathbb{P}} \mu = \mathbb{E}[X_1].$$

Theorem 1.4 (Central Limit Theorem). *If, in addition, $0 < \sigma^2 = \text{Var}(X_1) < \infty$, then*

$$\frac{\sqrt{n}(\overline{X}_n - \mu)}{\sigma} \Rightarrow \mathcal{N}(0, 1).$$

Thus $\overline{X}_n$ has approximate standard deviation $\sigma/\sqrt{n}$ for large n .

The LLN says that the estimation error vanishes; the CLT gives its $n^{-1/2}$ scale and approximate Gaussian shape.

1.4 Characteristic functions

Definition and basic properties. The characteristic function of X is

$$\varphi_X(u) = \mathbb{E}[e^{iuX}], \quad u \in \mathbb{R}.$$

It always exists because $|e^{iuX}| = 1$. Moreover,

$$\varphi_X(0) = 1, \quad |\varphi_X(u)| \leq 1, \quad \varphi_{aX+b}(u) = e^{iub} \varphi_X(au).$$

The characteristic function uniquely determines the law. If $\mathbb{E}[|X|^k] < \infty$, then it can be differentiated at the origin and

$$\varphi_X^{(k)}(0) = i^k \mathbb{E}[X^k].$$

Independent sums. If X and Y are independent, then

$$\varphi_{X+Y}(u) = \varphi_X(u) \varphi_Y(u).$$

This turns convolution of laws into multiplication of functions.

Formulas to know.

$$\begin{aligned} X \sim \text{Bernoulli}(p) : \quad & \varphi_X(u) = 1 - p + pe^{iu}, \\ X \sim \text{Poisson}(\lambda) : \quad & \varphi_X(u) = \exp(\lambda(e^{iu} - 1)), \\ X \sim \text{Exp}(\lambda) : \quad & \varphi_X(u) = \frac{\lambda}{\lambda - iu}, \\ X \sim \mathcal{N}(\mu, \sigma^2) : \quad & \varphi_X(u) = \exp\left(i\mu u - \frac{1}{2}\sigma^2 u^2\right). \end{aligned}$$

For $Z \sim \mathcal{N}(0, 1)$, differentiation under the integral followed by integration by parts gives

$$\varphi_Z'(u) = -u\varphi_Z(u), \quad \varphi_Z(0) = 1.$$

Hence $\varphi_Z(u) = e^{-u^2/2}$; writing $X = \mu + \sigma Z$ gives the general Gaussian formula.

1.5 Information, measurability, and conditional expectation

σ -algebras as information. A sub- σ -algebra $\mathcal{G} \subseteq \mathcal{F}$ represents the events that can be decided from the available information. A variable Z is $\mathcal{G}$ -measurable when its value can be determined from that information. The information revealed by a signal Y is $\sigma(Y)$.

On a finite space, $\mathcal{G}$ is generated by a partition $A_1, \dots, A_m$. A variable is $\mathcal{G}$ -measurable if and only if it is constant on every atom A_j . Equivalently,

$$Z \text{ is } \sigma(Y)\text{-measurable} \iff Z = g(Y)$$

for some measurable g .

Conditional probability and conditional laws. For events A, B with $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

For discrete variables and $\mathbb{P}(Y = y) > 0$,

$$\mathbb{P}(X = x \mid Y = y) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)}.$$

For a joint density and $f_Y(y) > 0$,

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}, \quad f_Y(y) = \int_{\mathbb{R}} f_{X,Y}(x, y) \, dx.$$

Definition of conditional expectation. For $X \in L^1$ and $\mathcal{G} \subseteq \mathcal{F}$, the conditional expectation $\mathbb{E}[X \mid \mathcal{G}]$ is the almost-surely unique random variable Z such that

$$Z \text{ is } \mathcal{G}\text{-measurable,} \quad \int_A Z \, d\mathbb{P} = \int_A X \, d\mathbb{P} \quad \text{for every } A \in \mathcal{G}.$$

It is the prediction of X based only on the information in $\mathcal{G}$. Conditioning on Y means conditioning on $\sigma(Y)$:

$$\mathbb{E}[X \mid Y] = \mathbb{E}[X \mid \sigma(Y)].$$

Finite-partition formula. If $\mathcal{G} = \sigma(A_1, \dots, A_m)$, where the A_j are partition atoms with positive probability, then

$$\mathbb{E}[X \mid \mathcal{G}] = \sum_{j=1}^m \frac{\mathbb{E}[X \mathbf{1}_{A_j}]}{\mathbb{P}(A_j)} \mathbf{1}_{A_j} = \sum_{j=1}^m \mathbb{E}[X \mid A_j] \mathbf{1}_{A_j}.$$

Conditional expectation is therefore the average of X inside each information set.

Formula for $\mathbb{E}[h(X) \mid Y]$: discrete case. Define, for every y with $\mathbb{P}(Y = y) > 0$,

$$\begin{aligned} m(y) &= \mathbb{E}[h(X) \mid Y = y] \\ &= \sum_x h(x) \mathbb{P}(X = x \mid Y = y) \\ &= \frac{\sum_x h(x) \mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)}. \end{aligned}$$

Then

$$\boxed{\mathbb{E}[h(X) \mid Y] = m(Y).}$$

Formula for $\mathbb{E}[h(X) | Y]$: density case. For $f_Y(y) > 0$, define

$$\begin{aligned} m(y) &= \mathbb{E}[h(X) | Y = y] \\ &= \int_{\mathbb{R}} h(x) f_{X|Y}(x | y) \, dx \\ &= \frac{\int_{\mathbb{R}} h(x) f_{X,Y}(x, y) \, dx}{\int_{\mathbb{R}} f_{X,Y}(x, y) \, dx}. \end{aligned}$$

Again,

$$\boxed{\mathbb{E}[h(X) | Y] = m(Y).}$$

The final substitution $y = Y$ is essential: a conditional expectation is a random variable, not merely a number indexed by y .

Essential conditional-expectation properties. For integrable variables, constants a, b , and $\mathcal{H} \subseteq \mathcal{G} \subseteq \mathcal{F}$,

$$\begin{aligned} \mathbb{E}[aX + bY | \mathcal{G}] &= a\mathbb{E}[X | \mathcal{G}] + b\mathbb{E}[Y | \mathcal{G}], \\ X \leq Y &\implies \mathbb{E}[X | \mathcal{G}] \leq \mathbb{E}[Y | \mathcal{G}], \\ Z \text{ } \mathcal{G}\text{-measurable} &\implies \mathbb{E}[ZX | \mathcal{G}] = Z\mathbb{E}[X | \mathcal{G}], \\ X \text{ } \mathcal{G}\text{-measurable} &\implies \mathbb{E}[X | \mathcal{G}] = X, \\ X \text{ independent of } \mathcal{G} &\implies \mathbb{E}[X | \mathcal{G}] = \mathbb{E}[X], \\ \mathbb{E}[\mathbb{E}[X | \mathcal{G}] | \mathcal{H}] &= \mathbb{E}[X | \mathcal{H}]. \end{aligned}$$

In particular,

$$\mathbb{E}[\mathbb{E}[X | \mathcal{G}]] = \mathbb{E}[X].$$

For convex ϕ , conditional Jensen gives

$$\phi(\mathbb{E}[X | \mathcal{G}]) \leq \mathbb{E}[\phi(X) | \mathcal{G}]$$

whenever the quantities are integrable.

Gaussian conditioning. If (X, Y) is jointly Gaussian, with standard deviations $\sigma_X, \sigma_Y > 0$ and correlation ρ , then

$$X | (Y = y) \sim \mathcal{N}\left(\mu_X + \rho \frac{\sigma_X}{\sigma_Y}(y - \mu_Y), \sigma_X^2(1 - \rho^2)\right).$$

Consequently,

$$\mathbb{E}[X | Y] = \mu_X + \rho \frac{\sigma_X}{\sigma_Y}(Y - \mu_Y), \quad \text{Var}(X | Y) = \sigma_X^2(1 - \rho^2).$$

The conditional mean is affine in the observed signal, and the residual variance decreases as $|\rho|$ increases.

1.6 Processes, filtrations, and martingales

Processes and filtrations. A discrete-time stochastic process is a sequence $(X_n)_{n \geq 0}$ of random variables on the same probability space. A filtration is an increasing family of σ -algebras,

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots,$$

representing information accumulated through time. The natural filtration of X is

$$\mathcal{F}_n^X = \sigma(X_0, \dots, X_n).$$

A process is adapted if X_n is $\mathcal{F}_n$ -measurable for every n . Adaptedness is the no-look-ahead requirement.

Random walks. If

$$S_n = S_0 + \sum_{k=1}^n \xi_k,$$

where the increments are i.i.d. with mean m and variance σ^2 , then

$$\mathbb{E}[S_n] = S_0 + nm, \quad \text{Var}(S_n) = n\sigma^2.$$

Martingales. An integrable adapted process (M_n) is a martingale if

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n \quad \text{for every } n.$$

For the random walk above,

$$M_n = S_n - nm$$

is a martingale. For independent symmetric ± 1 increments, $S_n^2 - n$ is also a martingale. A martingale combines three requirements: integrability, adaptation, and the conditional fair-game identity.

1.7 Differential-calculus review

One-variable derivative and Taylor expansion. For $g : \mathbb{R} \rightarrow \mathbb{R}$,

$$g'(x) = \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h}.$$

If g is twice differentiable near x , then

$$g(x+h) = g(x) + g'(x)h + \frac{1}{2}g''(x)h^2 + o(h^2).$$

Gradient and Hessian. For $f : \mathbb{R}^d \rightarrow \mathbb{R}$,

$$\nabla f(x) = \begin{pmatrix} \partial_1 f(x) \\ \vdots \\ \partial_d f(x) \end{pmatrix}, \quad \nabla^2 f(x) = (\partial_{ij} f(x))_{1 \leq i, j \leq d}.$$

The gradient gives first-order sensitivities; the Hessian records curvature and interactions. The second-order multivariable expansion is

$$f(x+h) = f(x) + \nabla f(x)^\top h + \frac{1}{2}h^\top \nabla^2 f(x)h + o(\|h\|^2).$$

Multivariable chain rule. If $x(t) \in \mathbb{R}^d$ and f are differentiable, then

$$\frac{d}{dt}f(x(t)) = \nabla f(x(t))^\top x'(t).$$

In particular, for $V = V(t, s)$ and a differentiable path $S(t)$,

$$\frac{d}{dt}V(t, S(t)) = V_t(t, S(t)) + V_s(t, S(t))S'(t).$$

The second-order Taylor term involving V_{ss} will be essential once the underlying path is Brownian rather than differentiable.

Lecture 1 checklist

You should be able to:

- pass between a law, a CDF, a density, and interval probabilities;
- compute $\mathbb{E}[h(X)]$, moments, variances, covariances, and variances of sums;
- state when dominated convergence allows a limit to pass through $\mathbb{E}$;
- distinguish the conclusions of the LLN and the CLT;
- compute and use characteristic functions, especially for independent sums;
- identify the information in $\sigma(Y)$ and test measurability on partition atoms;
- compute $\mathbb{E}[h(X) | Y]$ in both discrete and density models;
- use linearity, known factors, independence, monotonicity, Jensen, and the tower rule;
- apply the Gaussian conditioning formula;
- verify adaptation and the martingale property;
- compute derivatives, gradients, Hessians, Taylor expansions, and chain rules.

2 Lecture 2 — Brownian Motion, Quadratic Variation, and GBM

References

- *Stochastic Calculus for Finance II: Continuous-Time Models*, S. E. Shreve, Chapter 3, pp. 83-117
- *Brownian Motion and Stochastic Calculus*, I. Karatzas and S. E. Shreve, Chapter 2, pp. 47-126

2.1 From Discrete Random Walks to Brownian Motion

Simple symmetric random walk. Let $(\varepsilon_k)_{k \geq 1}$ be i.i.d. with $\mathbb{P}(\varepsilon_k = \pm 1) = \frac{1}{2}$, and define

$$S_0 = 0, \quad S_n = \sum_{k=1}^n \varepsilon_k.$$

Then (S_n) is a discrete-time martingale with independent increments, $\mathbb{E}[S_n] = 0$ and $\text{Var}(S_n) = n$.

Diffusive scaling. Introduce a time step $1/n$ and define the rescaled process

$$W^{(n)}(t) = \frac{1}{\sqrt{n}} S_{[nt]}, \quad t \geq 0,$$

with linear interpolation. By the CLT, increments of $W^{(n)}$ over intervals of length $t - s$ are approximately Gaussian with variance $t - s$.

Donsker invariance principle (informal). As $n \rightarrow \infty$, $W^{(n)}$ converges in distribution to a continuous-time process W , called *Brownian motion*. This result explains why Brownian motion is the universal continuous-time limit of random walks.

2.2 Definition and Basic Properties of Brownian Motion

Definition. A process $(W_t)_{t \geq 0}$ is a *standard Brownian motion* if:

- $W_0 = 0$,
- it has independent increments,
- for $0 \leq s < t$, $W_t - W_s \sim \mathcal{N}(0, t - s)$,
- paths are continuous almost surely.

Gaussian process viewpoint. Brownian motion is a centered Gaussian process fully characterized by

$$\text{Cov}(W_s, W_t) = \min(s, t).$$

In particular, for $s < t$,

$$W_t = W_s + (W_t - W_s),$$

where W_s and $W_t - W_s$ are independent.

Martingale property. With respect to its natural filtration $\mathcal{F}_t = \sigma(W_s : s \leq t)$, Brownian motion is a martingale:

$$\mathbb{E}[W_t \mid \mathcal{F}_s] = W_s, \quad s \leq t.$$

2.3 Markov and Strong Markov Properties

Markov property. Brownian motion is Markov: conditionally on W_s , the future is independent of the past, if $t \geq s$,

$$\mathbb{P}(W_t \in A \mid \mathcal{F}_s) = \mathbb{P}(W_t \in A \mid W_s).$$

Equivalently,

$$W_t \mid W_s = x \sim \mathcal{N}(x, t - s).$$

Strong Markov property. Let τ be a stopping time. Conditionally on $\mathcal{F}_\tau$, the shifted process

$$(W_{\tau+t} - W_\tau)_{t \geq 0}$$

is a Brownian motion independent of $\mathcal{F}_\tau$. This is crucial for hitting times, barriers, and American-style features.

2.4 Reflection Principle and Hitting Times

Running maximum. Define the running maximum

$$M_t = \sup_{0 \leq s \leq t} W_s,$$

and the hitting time of level a :

$$\tau_a = \inf\{t \geq 0 : W_t = a\}.$$

Reflection principle. For $a > 0$ and $t > 0$,

$$\mathbb{P}(M_t \geq a) = 2 \mathbb{P}(W_t \geq a).$$

More generally, for $w \leq a$,

$$\mathbb{P}(M_t > a, W_t < w) = \mathbb{P}(W_t > 2a - w).$$

Interpretation. Among paths that hit level a before time t , reflecting the path after the first hitting time preserves Wiener measure. This converts path-dependent events into Gaussian tail probabilities.

Consequence. Brownian motion hits every level almost surely:

$$\mathbb{P}(\tau_a < \infty) = 1, \quad a \in \mathbb{R}.$$

2.5 Pathwise Regularity and Irregularity

Nowhere differentiability. Almost surely, Brownian paths are nowhere differentiable.

Oscillations. For every $t \geq 0$, almost surely,

$$\limsup_{h \downarrow 0} \frac{W_{t+h} - W_t}{\sqrt{h}} = +\infty, \quad \liminf_{h \downarrow 0} \frac{W_{t+h} - W_t}{\sqrt{h}} = -\infty.$$

Hölder regularity. For every $\alpha < \frac{1}{2}$, Brownian motion is locally α -Hölder, but it is not $\frac{1}{2}$ -Hölder.

2.6 Quadratic Variation

Definition. For a partition $\pi = \{0 = t_0 < \dots < t_n = T\}$, define

$$[X]_T^\pi = \sum_{k=0}^{n-1} (X_{t_{k+1}} - X_{t_k})^2.$$

If this converges as $|\pi| \rightarrow 0$, the limit is the quadratic variation $[X]_T$.

Finite variation vs. Brownian motion. If $X_t = \int_0^t v_s ds$, then $[X]_T = 0$. In contrast, for Brownian motion,

$$[W]_T = T.$$

Key martingale. Quadratic variation explains why

$$W_t^2 - t$$

is a martingale. This is the simplest manifestation of Itô calculus.

Quadratic covariation. For processes X, Y ,

$$[X, Y]_T = \lim_{\pi} \sum_k (X_{t_{k+1}} - X_{t_k})(Y_{t_{k+1}} - Y_{t_k}).$$

If A has finite variation, then $[A, W] = 0$.

2.7 Heuristic Itô Rules

Brownian increments satisfy the scaling:

$$\Delta W = O(\sqrt{\Delta t}).$$

As a bookkeeping device,

$$(dW_t)^2 = dt, \quad dW_t dt = 0, \quad (dt)^2 = 0.$$

These rules encode which discrete sums survive in the limit.

2.8 Geometric Brownian Motion (GBM)

Motivation. Additive models allow negative prices and ignore scale effects. Instead, model relative changes:

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t.$$

Explicit solution. The solution is

$$S_t = S_0 \exp\left((\mu - \tfrac{1}{2}\sigma^2)t + \sigma W_t\right),$$

so S_t is lognormally distributed.

Moments.

$$\mathbb{E}[S_t] = S_0 e^{\mu t}, \quad \text{Var}(S_t) = S_0^2 e^{2\mu t} (e^{\sigma^2 t} - 1).$$

Volatility scaling. Log-returns over Δt have variance $\sigma^2 \Delta t$, which justifies the $\sqrt{\Delta t}$ rule.

2.9 Stopping Times and Optional Stopping

Stopped process. For a stopping time τ and a martingale (M_t) , define

$$M_t^\tau = M_{t \wedge \tau}.$$

Then (M_t^τ) is again a martingale.

Optional stopping (safe version). If τ is bounded, then

$$\mathbb{E}[M_\tau] = \mathbb{E}[M_0].$$

In general, $\tau < \infty$ a.s. is not sufficient; integrability or uniform integrability is required.

Example. For Brownian motion and $\tau_a = \inf\{t : W_t = a\}$, $\mathbb{E}[\tau_a] = \infty$ and $W_{\tau_a} = a$, so $\mathbb{E}[W_{\tau_a}] \neq \mathbb{E}[W_0]$. Optional stopping applies only to $t \wedge \tau_a$.

2.10 Multidimensional Brownian Motion

Definition. A d -dimensional Brownian motion is a vector-valued process

$$W_t = (W_t^1, \dots, W_t^d), \quad t \geq 0,$$

such that:

- $W_0 = 0 \in \mathbb{R}^d$,
- it has independent increments,
- for $0 \leq s < t$, the increment $W_t - W_s$ is multivariate Gaussian with mean 0 and covariance matrix $(t - s)I_d$,
- paths are continuous almost surely.

Equivalently, the components $(W^1, \dots, W^d)$ are independent one-dimensional Brownian motions.

Gaussian characterization. Multidimensional Brownian motion is a centered Gaussian process with

$$\text{Cov}(W_s^i, W_t^j) = \delta_{ij} \min(s, t), \quad 1 \leq i, j \leq d,$$

where δ_{ij} is the Kronecker symbol.

Independent vs. correlated Brownian motions. More generally, one may consider a Brownian motion with covariance matrix $\Sigma \in \mathbb{R}^{d \times d}$ (symmetric, positive semidefinite), defined by

$$\text{Cov}(W_s, W_t) = \min(s, t) \Sigma.$$

In this case,

$$d[W^i, W^j]_t = \Sigma_{ij} dt.$$

Any such process can be written as

$$W_t = B \widetilde{W}_t,$$

where $\widetilde{W}$ is a standard d -dimensional Brownian motion and B satisfies $BB^\top = \Sigma$ (e.g. Cholesky factorization).

Quadratic variation and covariation. For a d -dimensional Brownian motion W ,

$$[W^i]_t = t, \quad [W^i, W^j]_t = 0 \quad (i \neq j)$$

in the independent case, and more generally

$$[W]_t = t \Sigma,$$

where $[W]_t$ denotes the matrix-valued quadratic variation.

Itô rules (multidimensional mnemonic). For independent components,

$$(dW_t^i)^2 = dt, \quad dW_t^i dW_t^j = 0 \ (i \neq j), \quad dW_t^i dt = 0, \quad (dt)^2 = 0.$$

With correlation matrix Σ , this becomes

$$dW_t^i dW_t^j = \Sigma_{ij} dt.$$

Preview. In higher dimension, stochastic calculus keeps the same structure: *drift terms*, *martingale terms*, and *quadratic variation*. Only the bookkeeping becomes matrix-valued. This naturally leads to the multidimensional Itô formula and to Girsanov's theorem in vector form.